Dr. Huilin Huang

Assistant Professor · DEPARTMENT OF ENVIRONMENTAL SCIENCES

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Education

Ph.D in Geography, University of California, Los Angeles, USA *2021*
Committee: Professors Yongkang Xue (Chair), Dennis P. Lettenmaier, Kuo-Nan Liou, Gregory G. Okin, and Thomas Gillespie

B.S. in Atmospheric Sciences, Nanjing University, China *2015*

Professional Experience

2025-Present **Assistant Professor**, University of Virginia
2024-2025 **Earth Scientist**, Pacific Northwest National Laboratory
2021-2024 **Postdoctoral Researcher**, Pacific Northwest National Laboratory

Research Focus

- Wildfire and Land Cover Change
- Terrestrial Carbon Cycle and Biogeochemistry
- Land Surface Modeling and Land-Atmosphere Interaction

Grants

2026-2028	**From Prediction to Practice: A High-Resolution Fire-Response Framework to Enhance Community Resilience**, UVA Environmental Institute - PI	*$ 100,000*
2026-2028	**Next-generation forest fire modeling leveraging very-high-resolution remote sensing and deep learning**, UVA Environmental Institute - Co-PI	*$ 100,000*
Selectable	**Downscaling snow information to understand the impacts of scale and uncertain future climate on wildlife decision making**, NASA - Institutional-PI	*$ 1,199,999*
2022-2024	**Testing Hybrid Modeling Strategies for Machine-learning-based Fire Model in the Earth System Model**, DOE - Co-PI	*$ 161,000*
2024-Ongoing	**Developing an Integrated Climate-Ecosystem-Wildfire Model to Assess the Impact of Energy Systems on Wildfires**, DOE - Co-PI	*$ 177,000*

Grants Under Review

2026-2029 **Resilient Agriculture And Farming Communities in Southwestern California**, USDA- 2025 - Co-PI
2026-2029 **Fires at the Wildland-Urban Interface: Assessing Reproductive Effects (FLARE)**, NIH - 2025 - Co-PI
2025-2028 **Collaborative Research: FIRE-WUI: Closing the Loop: from Grid-induced Wildfire Prevention to Post-Wildfire Grid Restoration**, NSF Wildfire - 2025 - Co-PI
2025-2028 **Investigation of Extreme Weather Events Associated with Atmospheric Rivers and Their Impact on the Antarctic Ice Surface**, NSF 23-508 - Co-PI
2025-2028 **The Influence of Coupling Scale and Soil Moisture on Land-Atmosphere Interaction**, DOE-FOA - Co-PI

Publications

Huang H., Mcdowell, N., Chen, Y., et al.: Warming amplifies forest vulnerability to fire-induced canopy loss across the United States, In Preparation

Li, Z., **Huang H.**, et al.: Integrating ELM-BGC Simulated Carbon Cycling into ATS-PFLOTRAN Hydro-Biogeochemical Modeling in a Wildfire-Impacted Pacific Northwest Watershed, Environ. Res. Lett., In Review

Li, F., [including **Huang H.**], et al.: The Fire Modeling Intercomparison Project (FireMIP) contribution to CMIP7, Geosci. Model Dev., In Review

Shi, H., Huang, K., [including **Huang H.**], et al.: Cross-scale dominance of soil moisture in dryland vegetation growth, Nat. Clim. Change, In Review

Huang, H, Qian, Y., Hao, D., McDowell, N.G., Rogers, B., et al.: Elevated forest canopy loss after wildfires in moist and cool forests in the Pacific Northwest. Earth's Future, 13(7), p.e2025EF006373, 2025.

Liu, Y., **Huang, H.**, Wang, S. C., Zhang, T., Xu, D., et al: ML4Fire-XGBv1.0: Improving North American wildfire prediction by integrating a machine-learning fire model in a land surface model, Geosci. Model Dev., 4103-4117, 2025.

Huang, H, Qian, Y., Bisht, G., Wang, J., Chakraborty, T., et al.: WRF-ELM v1.0: a Regional Climate Model to Study Atmosphere-Land Interactions Over Heterogeneous Land Use Regions, Geosci. Model Dev., 18(5), 1427-1443, 2025

Shi, M., McDowell, N.G., **Huang, H.**, Zahura, F., et al: Ecosystem leaf area, gross primary production, and evapotranspiration responses to wildfires in the Columbia River Basin, Biogeosciences, 22(9), 2225-2238, 2025

Chen, Z., Lubis, S. W., Lu, J., Chang, C., **Huang, H.**, et al.: Eastward-shifting boreal summer intraseasonal oscillation amplifies North America heatwave and wildfire risks in warming climate. npj Clim. Atmos. Sci., 8(1), 317. 10.1038/s41612-025-01196-5, 2025

Park, C., Takahashi, K., Fujimori, S., Jansakoo, T., Burton, C., **Huang, H.**, et al.: Attributing human mortality from fire PM2.5 to climate change, Nat. Clim. Change, 10.1038/s41558-024-02149-1, 2024.

Burton, C., Lampe, S., Kelley, D., Thiery, W., Hantson, S., [including **Huang H.**], et al.: Global burned area increasingly explained by climate change, Nat. Clim. Change, 10.1038/s41558-024-02140-w, 2024.

Yang, Z., Wang, J., Qian, Y., Chakraborty, T., Xue, P., [including **Huang, H**], et al.: Future precipitation characteristic change and mechanisms over the Great Lakes Region through the Pseudo-global warming approach, J. Geophys. Res. Atmos, JGRD59684, 10.1029/2024JD041011, 2024.

Li, J., Zhang, K., Hassan, T., Zhang, S., Ma, P. L., [including **Huang, H**], et al.: Assessing the sensitivity of aerosol mass budget and effective radiative forcing to horizontal grid spacing in E3SMv1 using a regional refinement approach, Geosci. Model Dev., 17, 1327-1347, 10.5194/gmd-17-1327-2024, 2024.

Hao, D., Bisht, G., Wang, H., Xu, D., **Huang, H**, Qian, Y., and Leung, L. R.: A cleaner snow future mitigates Northern Hemisphere snowpack loss from warming, Nat. Commun., 14, 6074, 10.1038/s41467-023-41732-6, 2023.

Liu, Y., Feng, S., Qian, Y., **Huang, H**, and Berg, L. K.: How do North American weather regimes drive wind energy at the sub-seasonal to seasonal timescales? npj Clim. Atmos. Sci., 6, 100, 2023.

Hao, D., Bisht, G., Rittger, K., Bair, E., He, C., **Huang, H**, et. al.: Improving snow albedo modeling in the E3SM land model (version 2.0) and assessing its impacts on snow and surface fluxes over the Tibetan Plateau, Geosci. Model Dev., 16, 75-94, 10.5194/gmd-16-75-2023, 2023.

Huang, H, Qian, Y., Liu, Y., He, C., Zheng, J., Zhang, Z., and Gkikas, A.: Where does the dust deposited over the Sierra Nevada snow come from?, Atmos. Chem. Phys., 22, 15469-15488, 10.5194/acp-22-15469-2022, 2022.

Huang, H, Qian, Y., He, C., Bair, E., and Rittger, K.: Snow albedo feedbacks enhance snow impurity-induced radiative forcing in the Sierra Nevada. Geophys. Res. Lett., e2022GL098102, 2022.

Huang, H, Fischella, M. R., Liu, Y., Ban, Z., Fayne, J. V., Li, D., Cavanaugh, K. C., and Lettenmaier, D. P.: Changes in mechanisms and characteristics of western U.S. floods over the last sixty years. Geophys. Res. Lett, 49, e2021GL097022. https://doi.org/10.1029/2021GL097022, 2022.

Huang, H, Xue, Y., Liu, Y., Li, F., and Okin, G. S.: Modeling the short-term fire effects on vegetation dynamics and surface energy in southern Africa using the improved SSiB4/TRIFFID-Fire model, Geosci. Model Dev., 14, 7639-7657, 10.5194/gmd-14-7639-2021, 2021.

Liu, Y., W. Guo, **Huang, H**, J. Ge, and B. Qiu. Estimating global aerodynamic parameters in 1982–2017 using remote-sensing data and a turbulent transfer model, Remote Sens. Environ., 260, 112428, 2021

Xue, Y., Yao, T., Boone, A. A., Diallo, I., Liu, Y., [including **Huang, H**], et al.: Impact of Initialized Land Surface Temperature and Snowpack on Subseasonal to Seasonal Prediction Project, Phase I (LS4P-I): organization and experimental design, Geosci. Model Dev., 14(7), 4465-4494, 10.5194/gmd-14-7639-2021, 2021.

Huang, H, Xue, Y., Li, F., and Liu, Y.: Modeling long-term fire impact on ecosystem characteristics and surface energy using a process-based vegetation–fire model SSiB4/TRIFFID-Fire v1.0, Geosci. Model Dev., 13, 6029-6050, 10.5194/gmd-13-6029-2020, 2020.

Huang, H, Xue, Y., Chilukoti, N., Liu, Y., Chen, G., and Diallo, I.: Assessing global and regional effects of reconstructed land use and land cover change since 1950 on climate using a coupled land-atmosphere-ocean model, J. Clim., 10.1175/JCLI-D-20-0108.1, 2020.

Huang, H, Gu, Y., Xue, Y., Jiang, J., et al: Assessing aerosol indirect effect on clouds and regional climate of East/South Asia and West Africa using NCEP GFS, Clim. Dyn., 52(9-10), 5759-5774, 2019.

Wanders, N., Bachas, A., He, X. G., **Huang, H**, Koppa, A., et al.: Forecasting the Hydroclimatic Signature of the 2015/16 El Nino Event on the Western United States, J. Hydrometeorol., 18, 177-186, 10.1175/Jhm-D-16-0230.1, 2017.

Presentations

Selected Oral Presentations

(Invited) Huang, H: Understanding the Burning Questions: Wildfire Dynamics and Interactions with Climate, Ecosystems, and Human. Nov. 2025, Climate and Global Dynamics Laboratory, NCAR, Boulder, CO

(Invited) Huang, H: Understanding the Burning Questions: Wildfire Dynamics and Its Impact on Ecosystems and Climate. Sep. 2025, University of Virginia, Charlottesville, VA

(Invited) Huang, H: Understanding the Burning Questions: Wildfire Dynamics and Its Impact on Ecosystems and Climate. Feb. 2025, University of Virginia, Charlottesville, VA

(Invited) Huang, H: Understanding the Burning Questions: Wildfire Dynamics and Its Impact on Ecosystems and Climate. Feb. 2025, Texas A & M University, College Station, TX

(Invited) Huang, H: Land-Atmosphere Interactions: Insights into Environmental Change. Feb. 2025, UC Merced, Merced, CA

Huang, H., Qian, Y., Rittger, K., Jiang, P., and Bair, E.: High-Resolution Assessment of Climate Change Impacts on Slope-Scale Snow in the Sierra Nevada. Dec. 2024, American Geophysical Union (AGU) Annual Meeting, Washington, D.C.

Li, Z., **Huang, H.**, Xiao, Y., Li, B., Hammond, G., Chen, X.: Integrating ELM-BGC Carbon Cycling with ATS-PFLOTRAN Hydro-Biogeochemical Modeling in a Wildfire-Impacted Pacific Northwest Watershed. Dec. 2024, AGU Annual Meeting, Washington, D.C.

(Invited) Huang, H: Understanding Interactions between Tree Canopy and Fire in the Pacific Northwest in a Changing Climate, Sep. 2024, Wildfire Research Summit, PNNL, Richland, WA

(Invited) Huang, H: Understanding the Burning Questions: Fire Impacts on Ecosystems Through Observations and Modeling, Aug. 2024, Stanford University, Stanford, CA

(Invited) Huang, H: Understanding the Burning Questions: The History of Wildfire and its Impacts on Ecosystem, Energy, and Climate. March 2024, Colorado State University, Fort Collins, CO

Huang, H., Qian, Y., Bisht, G., Wang, J., Chakraborty, T., Hao, D., Li, J., Thurber, T., Singh, B., Yang, Z., Liu, Y., Xue, P., Sacks, W., Coon, E., and Hetland, R.: The Relevance of Coupled Climate Model WRF-ELM for Atmosphere-Coastal-Urban-Rural Interaction Analysis. Jan. 2024, American Meteorological Society (AMS) Annual Meeting, Baltimore, MD

(Invited) Huang, H: Understanding the Burning Questions: The Past, Present, and Future of Wildfires and the Impacts on Ecosystem and Climate. Dec. 2023, Missouri State University, Springfield, MO

Huang, H., Qian, Y., Liu, Y., He, C., Zheng, J., Zhang, Z., and Gkikas, A.: Where Does the Dust Deposited over the Sierra Nevada Snow Come From? Dec. 2022, AGU Annual Meeting, Chicago, IL

(Invited) Huang, H: Snow impurities induce early floods in the Sierra Nevada. Center for Western Weather and Water Extremes (CW3E). July 2022, University of California, San Diego, CA

Huang, H., Qian, Y., He, C., Bair, E., and Rittger K.: Evaluation of Snowpack Simulation and the Role of Impurity Effects in the Sierra Nevada based on High-resolution Regional Climate-aerosol-snow Modeling. Dec. 2021, AGU Annual Meeting, New Orleans, LA

Huang, H., Xue, Y., Chilukoti, N., Liu, Y., Chen, G., and Diallo, I.: Assessing global biogeophysical effects of reconstructed land use and land cover change on climate since 1950 using a coupled land–atmosphere–ocean model. April 2021, European Geophysical Union (EGU) Annual Meeting, Vienna, Austria

(Invited) Huang, H, Xue, Y., Li, F., and Liu, Y.: Modeling fire impact on vegetation dynamics at seasonal and interannual scales using SSiB4/TRIFFID-Fire. Oct. 2020, Fire Model Intercomparison Project (FIREMIP) Workshop, Beijing, China

Huang, H., Xue, Y., Chilukoti, N., Liu, Y., and Chen, G.: Assessing global and regional effects of reconstructed land use and land cover change since 1950 on climate using a coupled land-atmosphere-ocean model. Jan. 2020, AMS Annual Meeting, Boston, MA

Huang, H., Xue, Y., Chilukoti, N., Liu, Y., and Chen, G.: Assessing global and regional effects of reconstructed land use and land cover change since 1950 on climate using a coupled land-atmosphere-ocean model. Dec. 2019, AGU Annual Meeting, San Francisco, CA

(Invited) Huang, H, Oaida, C., and Xue, Y.: Modeling aerosols in snow in SSiB. Jul. 2019, 2nd International GEWEX/GASS/LS4P and TPEMIP Workshop, Nanjing, China

Huang, H., Gu, Y., Xue, Y., Jiang, J. H., and Zhao, B.: Assessing aerosol indirect effects on clouds and regional climate of East/South Asia and West Africa using NCEP GFS. Dec. 2018, AGU Annual Meeting, Washington, D.C.

Selected Poster Presentations - Presenting Author

Huang, H., Qian, Y., Hao, D., McDowell, N.G., Rogers, B., Li, L., Shi, M., Rittger, K., Bisht, G., and Chen, X.: Elevated forest loss after wildfires in moister and cooler forests in the Pacific Northwest, Dec. 2024, AGU Annual Meeting, Washington, D.C.

Huang, H., Qian, Y., Li, L., Shi, M., Hao, D., Bisht, G., and Chen, X.: Fire Impacts on the Surface Albedo and Water Cycle in the Columbia River Basin, Dec. 2023, AGU Annual Meeting, San Francisco, CA

Huang, H., Xue, Y., Li. F., Liu, Y.: Seasonal fire impact on ecosystem characteristics and surface energy in Southern Africa savanna. Dec. 2020, AGU Annual Meeting, San Francisco, CA

Huang, H., Gu, Y., Xue, Y., Lu, C., and Jiang, J. H.: Assessing aerosol indirect effects on clouds and regional climate of East/South Asia and West Africa using NCEP GFS. Dec 2017, AGU Annual Meeting, New Orleans, IL

Huang, H., Gu, Y., Xue, Y., and Lu, C.: Assessing the aerosol direct and first indirect effects using ACM/GCM simulation results. Dec 2016, AGU Annual Meeting, San Francisco, CA

Research Experience

Pacific Northwest National Laboratory

- Predict seasonal wildfire risk on energy infrastructure and electricity resiliency, in the project ”North American Energy Resilience Model” (2024-present)
- Assess the effects of meteorology and ocean wave conditions for realistic wind predictions, in the project ”Addressing Challenges in Energy: Floating Wind in a Changing Climate (ACE-FWICC)” (PI-Larry Berg at PNNL) (2024-present)
- Integrated a machine-learning-based fire model in the Energy Exascale Earth System Model (E3SM) based on a hybrid modeling approach. The model is further improved by incorporating powerline ignitions. (PI-Ye Liu at PNNL) (2022-present)
- Couple land model ELM to WRF (WRF-ELM) and examined land-atmosphere-lake coupling over the Great Lakes in the project ”COMPASS-Coastal Observations, Mechanisms, and Predictions Across Systems and Scales” (PI-Robert Hetland at PNNL) (2022-present)
- Investigated fire impact in the Pacific Northwest watersheds using satellite observations in the project ”River Corridor Hydrobiogeochemistry from Reaction to Basin Scale” (PI–Timothy Scheibe at PNNL) (2022-2024)
- Evaluated snow impurity effects and identified the sources of impurity using machine learning techniques in the NASA project ”The lifecycle of snow in the Sierra Nevada USA: from snowfall to snowmelt and effects on endangered bighorn sheep” (PI-Edward Bair at UCSB) (2021-2022)

UCLA Geography Department, Prof. Yongkang Xue’s group

- Developed a process-based fire model within the dynamic vegetation model (SSiB4/TRIFFID-Fire) and analyzed the long-term and short-term fire impact (2018-2021)
- Participated in the Fire Modeling Intercomparison Project (FireMIP) for future fire prediction (2020-present)
- Evaluated land use and land cover change (LULCC) on climate using global climate model (NCEP/CFS) (2017-2020)
- Assessed aerosol direct and indirect effects on regional climate using global climate model (NCEP/CFS) (2015-2017)

UCLA Geography Department, Prof. Dennis Lettenmaier’s group

- Investigating changes in flood mechanisms and intensity of the Western U.S. floods (2018-2021)

Teaching Experience

2019-2020 **People and Earth’s Ecosystems (50 undergrad students)**, Teaching Assistant
2018-2020 **Introduction to Biophysical Modeling of Land Surface Processes (12 grad students)**, Teaching Assistant
2021 **Human Impact on Biophysical Environment (110 undergrad students)**, Teaching Assistant

Mentoring

2018-2020	**Min-Ying Li, Yuan Ma, Ziwei Li**, Undergraduate student	*UCLA*
2020-2021	**Khalil Ganem, Zhijiong Cao**, Graduate Student	*UCLA*

Fellowship & Student Grants

2019	**Graduate Division Travel Grants**, UCLA
2018, 2019	**Geography Graduate Student Travel Grants**, UCLA
2016, 2020	**Departmental Fellowship**, UCLA

Outreach & Professional Development

Community Services

Associate Editor: Geophysical Research Letters, AGU, 2025-present

Early Career Editorial Board: AGU Advances, AGU, 2025-present

Panel Reviewer: NASA Research Programs, 2025

Strategy Team: Ecosystem Impacts, FireSat Earth Fire Alliance (2025–present)

Advisory Board: WCRP-CMIP Fire Model Intercomparison Project (2024 - present)

Review Committee: NOAH-MP land surface model biogeochemical component (May 2023 - present)

Committee Board: PNNL Wildfire Communities of Interest Network (COIN) (2024 - present)

Review Editor: Atmosphere and Climate section in Frontiers (IF 5.411: July 2023 - present)

Guest Editor: Special Issue: Advancing our Understanding of Wildfires: Mapping, Modeling, and Impact Assessment in Geomatics, Natural Hazards and Risk (IF 3.110: Dec. 2025-Dec 2026)

Guest Editor: Special Issue: Land–Atmosphere Coupling under Climate Change in Atmosphere (IF 3.110: Oct. 2022-April 2023)

Session Convener: Fire-Ecosystem Interaction in a Changing Environment (AGU 2025); World on Fire: Understanding Fire Impacts on Energy, Water, and Carbon Cycles, and Ramifications to Climate, Soil, and Ecosystem Structure and Functioning (AGU 2023, 2024); 22nd Symposium on the Coastal Environment (AMS 2024)

Peer Reviewer: Atmospheric Chemistry and Physics; Journal of Climate; Earth System Science Data; Earth's Future; Geoscientific Model Development; Geophysical Research Letter; Journal of Hydrology; Climate Change; Science of the Total Environment; Natural Hazards and Earth System Sciences; Atmosphere

Inclusive Services

Chair of Young Scientists Professional Development (YSPD): PNNL (April 2023 - present)

Student Representative for Hydrology Search Committee: UCLA Geography Department (Oct. 2019-Feb. 2020)

Regional Director: UCLA COAA-SCC, a non-profit organization with goals to facilitate professional networking among colleagues in Southern California (Aug. 2017-Aug. 2019)

Relevant Skills

Numerical Modeling (* denotes developers' level): *Energy Exascale Earth System Model (E3SM), *NCEP Climate Forecast System Version 2 (CFSv2), *The Weather Research and Forecasting (WRF) model, *Simplified Simple Biosphere Model (SSiB), The Weather Research and Forecasting model coupled with Chemistry (WRF-Chem), Variable Infiltration Capacity (VIC) model

Machine Learning: Deep Neural Network (DNN), Extreme Gradient Boosting (XGBoost), Random Forest, SHapley Additive exPlanations (SHAP)

Programming Languages and Tools: Python, Fortran, NCL, MATLAB, GrADS, C, ArcGIS, ENVI, QGIS, PhotoScan